

THOTH – تحوت

Autonomous Interactive Robot for Smart Museum Guidance

Task 1 Report

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1. Introduction

Museums worldwide are under increasing pressure to modernise visitor experiences in response to rising global tourism, diverse multilingual audiences, and growing expectations for digital engagement. Traditional static systems — audio wands, printed labels, and human guides — are proving insufficient to meet these demands at scale.

This report fulfils Task 1 of the THOTH graduation project. It is structured into three parts: (1) a technology survey cataloguing the electronic guidance systems currently deployed across real museums worldwide, categorised by function; (2) a literature review of recent academic work (2020–2025) that informs the technical foundation of THOTH; and (3) the aims, objectives, and justification for the proposed robotic system.

2. Electronic Technologies in Museums: A Global Survey

The following survey catalogues real deployments of visitor-facing electronic technologies across major museums worldwide. Technologies are grouped by function as defined in the task specification.

2.1 Audio & Multimedia Guides

Audio guides represent the most widely deployed form of electronic museum assistance, ranging from simple handheld wands to fully integrated smartphone platforms.

Technology	Description	Museum Example(s)
Traditional Audio Wands	Handheld devices; visitors input exhibit numbers for pre-recorded commentary	Egyptian Museum Cairo, many regional history museums
Smartphone Apps (BYOD)	iOS/Android apps offering audio tours, digital maps, and exhibit information	Bloomberg Connects App — British Museum, Guggenheim, Art Institute of Chicago
Multimedia Tablet Guides	iPads/tablets rented at front desk; feature video, 3D model rotations, and high-res images	Smithsonian National Museum of Natural History (tablet loan program)
GPS/Location-Aware Audio	Bluetooth or GPS triggers audio automatically when visitor enters exhibit zone	Louvre (beacon-triggered audio), Natural History Museum London

2.2 Augmented & Virtual Reality (AR/VR)

Immersive technologies are increasingly adopted in major institutions to reconstruct historical contexts or animate exhibits that cannot otherwise be experienced.

Technology	Description	Museum Example(s)
AR Mobile Apps	Phone camera overlays historical imagery, colour reconstruction, or contextual info onto physical exhibits	Civilisations AR App — British Museum, National Museum of Australia
AR Headsets (HoloLens)	Projects 3D holograms onto physical space, e.g., animated skeletons beside fossil exhibits	Smithsonian (HoloLens dinosaur project), Museum of Natural Sciences Brussels
VR Stations	Stationary headset pods transporting visitors to historical periods or reconstructed sites	GEM (Grand Egyptian Museum) VR rooms; Musée d'Orsay Paris VR experience

2.3 Interactive Exhibits & Displays

Interactive physical interfaces are now standard in major institutions, enabling multi-user engagement and deeper content exploration.

Technology	Description	Museum Example(s)
Interactive Touchscreens/Kiosks	Wall-mounted or freestanding screens for browsing collections, digital books, and floor maps	GEM (Grand Egyptian Museum), Louvre Abu Dhabi, Tate Modern London
Interactive Tables	Large horizontal multi-touch surfaces for collaborative map exploration or digital puzzles	Museum of London (multi-touch table), Science Museum London
Gesture-Based Technology	Screens responding to hand waves or body movement to control content	Ontario Science Centre, Exploratorium San Francisco
Tangible Interfaces	Physical replica objects placed on sensor pads to trigger related digital content	Natural History Museum London (fossil replica pads)

2.4 Accessibility & Inclusivity Technologies

Accessibility technology has moved from an optional add-on to a core requirement, with most leading institutions now offering multiple parallel access channels.

Technology	Description	Museum Example(s)
Assistive Listening Devices (ALD)	T-coil induction loop systems compatible with hearing aids	Metropolitan Museum of Art (Met), Tate Britain
Video Sign Language Guides	Pre-recorded BSL/ASL interpreter videos for specific exhibits	British Museum (BSL video guides), Victoria & Albert Museum
High-Contrast & Large-Print Tablets	Devices specifically configured for low-vision visitors	Smithsonian Accessibility Program, Science Museum London

Technology	Description	Museum Example(s)
Audio Description Devices	Dedicated tracks describing visual elements of artworks for blind visitors	National Gallery London, MoMA New York

2.5 Digital Navigation & Wayfinding

Indoor navigation remains a significant challenge globally — a finding strongly supported by the team's own user survey, in which 80% of respondents reported feeling lost or confused in museums.

Technology	Description	Museum Example(s)
Interactive Digital Signage	Large searchable directory screens for locating exhibits, facilities, and services	GEM, Louvre, British Museum entrance halls
Indoor Mapping / Beacon Technology	Bluetooth BLE beacons connected to museum apps providing turn-by-turn indoor navigation	Louvre (iBeacon system), MoMA, American Museum of Natural History
Queuing & Crowd Sensing	Real-time app features showing wait times and crowd density for popular exhibits	Vatican Museums (virtual queuing), Rijksmuseum Amsterdam

2.6 On-Demand Information & AI Labeling

Technology	Description	Museum Example(s)
QR Codes & NFC Tags	Scannable stickers next to labels linking to additional web content, curator notes, or videos	Victoria & Albert Museum, National Museum of Egyptian Civilization (NMEC)
Digital Label Systems	E-ink or LCD screens beside artifacts remotely updatable without physical reprinting	Tate Modern, Science Museum Group
Chatbots & AI Assistants	In-app or web AI answering visitor questions (hours, locations, exhibit info)	Smithsonian (Ask a Curator AI), Cooper Hewitt Design Museum chatbot

2.7 Personalization & Social Features

Technology	Description	Museum Example(s)
"My Museum" Profiles	App functionality allowing visitors to favourite exhibits and receive follow-up content via email	Art Institute of Chicago app, Cleveland Museum of Art (ArtLens)

Technology	Description	Museum Example(s)
Digital Scavenger Hunts / Gamification	App or tablet-based games encouraging exploration, targeted at families and school groups	Natural History Museum London (Dino Safari), Smithsonian Kids
AR Photo Filters & Social Sharing	AR lenses or digital frames encouraging social media sharing from within the museum	Musée d'Orsay Instagram filters, various modern art museums

2.8 Robotic Guide Systems (Direct Precedents for THOTH)

A small but growing category of museums have trialled or deployed autonomous robot guides. These represent the closest direct precedents for the THOTH system.

System / Robot	Capabilities	Deployment
Pepper Robot (SoftBank)	Social robot; speech interaction, screen display, limited mobility. Deployed as greeter and information assistant.	Various museums and public spaces globally including Musée de la Romanité, France
CLIO (Research System, 2024)	Full autonomous navigation (ROS2/Nav2), LLM-scripted tour narration, exhibit pointing with laser, face tracking, co-speech gestures. Evaluated in a live museum setting.	Academic research deployment — direct technical precedent (see Section 3)
Stevie (Trinity College Dublin)	Socially assistive robot; conversational interaction with elderly and museum visitors. Deployed for limited engagement studies.	Smithsonian American Art Museum pilot (2019–2020)
Paul (Städel Museum, Frankfurt)	Autonomous guide robot; greeting visitors, providing directions, basic exhibit information.	Städel Museum Frankfurt — operational deployment

3. Literature Review (2020–2025)

This section reviews four key academic works that directly inform the design decisions for THOTH. Each paper is assessed for its specific contribution to the project.

3.1 Velentza et al. (2020) — Robot Personality and Visitor Learning

Reference

Velentza, A.-M., Heinke, D., & Wyatt, J. (2020). Museum robot guides or conventional audio guides? An experimental study. *Advanced Robotics*.

Key Findings

- Visitors guided by a robot retained significantly more information than those using conventional audio guides.

- A robot's personality (cheerful vs. serious) directly impacts visitor learning outcomes and engagement levels.
- The physical, social presence of a robot creates higher attention and interaction compared to passive audio systems.

Relevance to THOTH

This paper provides the primary experimental justification for choosing a robotic system over an audio guide or mobile application. It directly supports Objective 2 (AI-Based Voice Interaction) and informs the interaction design philosophy: THOTH should exhibit a consistent, positive personality to maximise visitor learning outcomes.

3.2 CLIO (2024) — Full Tour Guide Robot with Co-Speech Actions

Reference

Grady et al. (2024). CLIO: A Tour Guide Robot with Co-Speech Actions for Visual Attention Guidance and Enhanced User Engagement. arXiv:2512.05389.

System Overview

CLIO is a state-of-the-art autonomous museum tour guide robot and represents the most technically complete published system directly comparable to THOTH. It was evaluated in a live museum environment and integrates several advanced capabilities:

- Navigation: FAST-LIO2 for 3D point cloud mapping; ROS2 Nav2 framework with A* path planning and MPPI controller for dynamic obstacle avoidance at 10 Hz.
- Tour Scripting: An LLM (OpenAI) parses raw tour scripts into a structured action queue, segmenting narration into sentences and associating each with concurrent robot actions (e.g., LookAtExhibit, PointLaser).
- Exhibit Detection: A YOLOv11 model trained on exhibit photographs determines 3D exhibit locations using an RGBD camera, enabling the robot to physically point at and look toward specific artifacts.
- Visitor Tracking: MediaPipe face detection combined with a Kalman filter enables the robot to track visitor head position at ~10 Hz, maintaining natural gaze during interaction.
- Co-speech Actions: Synchronised laser pointing, head turning toward exhibits, and natural eye blinking (23.3 blinks/min) contribute to visitor engagement and perceived robot intelligence.

Key Findings

- Combining gaze direction and laser pointing significantly outperforms either action alone in directing visitor visual attention.
- LLM-generated tour scripts with transitional narratives improve perceived tour flow and visitor engagement.
- Dynamic visitor tracking via face detection creates substantially more natural interaction than static forward-facing posture.

Relevance to THOTH

CLIO serves as the primary technical reference architecture for THOTH. Several design decisions are directly informed by this work: (1) the use of ROS2/Nav2 for navigation is validated as suitable for museum environments; (2) LLM-based tour scripting is adopted as the content delivery approach; (3) the importance of co-speech physical actions for

engagement supports THOTH's screen and audio output design; (4) the FAST-LIO2 + static map approach informs the navigation strategy discussion (see system design phase).

4. User Survey Analysis

A visitor preference survey was conducted to validate the need for an intelligent robotic guide system and to identify priority features. The survey complements the technology audit in Section 2 by capturing the user perspective.

4.1 Key Findings

- Navigation pain point confirmed: 80% of respondents reported feeling lost or confused inside museums — the single clearest signal for the value of THOTH's navigation guidance capability.
- Content preference: 73.3% preferred learning through videos and 3D models; 51.1% through interactive screens — validating THOTH's multimedia display as a core feature rather than an optional add-on.
- Robotic guidance acceptance: 48.9% would use a robot guide (Yes) and 48.9% responded Maybe — a combined 97.8% showing no strong rejection of robotic systems.
- Robot vs. mobile app: 73.3% believed a robot guide is better than a mobile app, citing physical guidance, engagement, and the ability to handle groups without requiring personal devices.
- Top desired features: AI question answering (82.2%), audio descriptions (86.7%), and navigation to exhibits (71.1%) — all core THOTH capabilities.
- Technology already familiar to respondents: Interactive touchscreens (68.9%) and digital maps (64.4%) are the most commonly used museum technologies, indicating users are comfortable with digital interfaces.

4.2 Survey Conclusion

The survey data strongly supports the THOTH concept. Navigation failure (80%), high acceptance of robotic guidance (97.8%), and the preference for interactive multimedia content (73.3%) collectively validate the system's three core pillars: autonomous navigation, AI interaction, and multimedia display. These findings are used directly to prioritise system features in the objectives defined in Section 5.

5. Problem Identification

Based on the combined evidence from the technology survey (Section 2), literature review (Section 3), and user survey (Section 4), the following problems in current museum guidance systems are identified:

Problem	Evidence	Impact
Navigation failure	80% of survey respondents felt lost; no widely deployed low-cost indoor navigation solution exists	Visitors miss exhibits, leave early, have negative experience
Passive, one-way information delivery	Audio guides used by <50% of respondents; 42.2% found no improvement in experience	Low engagement, poor knowledge retention (Velentza et al., 2020)
Human guide scalability limits	Guides cannot serve large multilingual groups simultaneously	Inconsistent visitor experience; high operational cost
Limited accessibility	97.8% of respondents agreed accessibility technologies are needed; current implementations are fragmented	Exclusion of hearing-impaired and low-vision visitors
No real-time dynamic assistance	Static systems cannot answer unexpected questions; AI chatbots are screen-bound only	Visitor questions go unanswered; engagement lost

6. Aims and Objectives

6.1 Main Aim

To develop an autonomous interactive robotic guide (THOTH) that enhances museum visitor experiences through physical navigation guidance, real-time AI-powered interaction, and multilingual multimedia content delivery — operating reliably in the real-world conditions of a large cultural institution.

6.2 Objectives

Objective 1: Autonomous Navigation

Enable the robot to physically guide visitors through museum spaces along predefined and dynamic routes, avoiding obstacles in real time.

Reason: 80% of surveyed visitors reported navigation confusion. No existing static technology (apps, signs, kiosks) provides physical guidance. CLIO (2024) and other robotic systems demonstrate this is technically achievable with ROS2/Nav2. Physical presence guidance is the core differentiator of a robot vs. a mobile app.

Objective 2: AI-Based Voice Interaction

Allow visitors to ask questions in natural spoken language and receive contextually accurate, conversational responses about specific exhibits.

Reason: Velentza et al. (2020) demonstrated that interactive robot guides produce significantly better knowledge retention than audio guides. 82.2% of survey respondents listed AI question answering as a desired feature. LLM-based interaction (as demonstrated by CLIO) enables responses to open-ended questions that static systems cannot handle.

Objective 3: Multimedia Content Delivery

Present exhibit-related images, videos, and visual information through an onboard display during tours and question-answering interactions.

Reason: 73.3% of respondents preferred learning through videos and 3D models — the highest preference of any content format. Text and audio alone are insufficient for visual exhibits. The display also enables accessibility features such as real-time captions. CLIO confirms screen-based output is a standard component in modern museum robot architectures.

Objective 4: Multilingual Support

Deliver guided content and conversational responses in at least Arabic and English, with architecture supporting additional language expansion.

Reason: Egypt's major museums — particularly the Grand Egyptian Museum — serve large international and local Arabic-speaking audiences. Human guides are limited in language coverage. Unlike mobile apps, a robot guide can serve visitors without requiring them to download software or own a compatible device.

Objective 5: Accessibility Features

Provide real-time caption display, adjustable audio volume, and screen-based information as accessible alternatives to audio-only content.

Reason: 97.8% of survey respondents agreed that museums should provide accessibility technologies. 86.7% specifically requested audio descriptions for exhibits. Current robotic systems such as CLIO do not address hearing-impaired visitors; THOTH can do so through its onboard display.

Objective 6: Real-Time Question Answering

Enable the robot to respond dynamically and accurately to visitor questions beyond the scripted tour content, using an LLM grounded in a curated knowledge base of museum exhibits.

Reason: Static tour scripts cannot anticipate every visitor question. CLIO's LLM-scripted architecture shows that language model integration in museum robots is viable. Grounding the LLM with exhibit-specific knowledge (Retrieval-Augmented Generation) prevents hallucination — a critical concern identified in the literature.

7. Justification: Why a Robot Rather Than a Mobile Application

A frequently raised question is whether an autonomous robot is genuinely necessary when a well-designed smartphone application could deliver audio guides, maps, and exhibit information at far lower cost. This section provides a structured answer grounded in survey data, literature, and engineering reasoning.

Dimension	Mobile Application	THOTH Robot
Physical Guidance	Cannot physically guide — user follows on-screen map	Robot physically leads visitor through the museum
Interaction Model	One-way or text-based; requires user to initiate on screen	Bidirectional spoken dialogue; robot proactively engages
Group Handling	Each visitor needs own device; no group coordination	One robot serves a group simultaneously
Device Dependency	Requires visitor to own a compatible smartphone with data/WiFi	Fully self-contained; no visitor hardware required
Engagement	Passive; visitors check phone intermittently	Active physical presence maintains attention (Velentza 2020)
Navigation Failure (survey)	Cannot solve the physical wayfinding problem	Directly addresses the 80% of visitors who feel lost
Accessibility	Screen-based only; no hands-free option	Audio + visual + physical direction in parallel
Knowledge Retention	Comparable to audio guide (low)	Significantly higher (Velentza et al., 2020)

The survey reinforces this analysis: 73.3% of respondents stated a robot is better than a mobile app, with qualitative responses emphasising physical presence, independence from personal devices, and the social engagement factor. The literature confirms that a robot's physical, embodied presence — not merely the information it delivers — is itself a significant driver of visitor engagement and learning outcomes.

8. Conclusion

This report has established three complementary evidence bases for the THOTH project. The technology survey identified the current state of museum guidance systems across seven functional categories, confirming that robotic guide systems remain rare and represent a genuine gap in deployed technology. The literature review — anchored by Velentza et al. (2020) and the CLIO system (2024) — provides validated technical approaches and empirical evidence for the effectiveness of robotic museum guides. The user survey confirms that the problems THOTH addresses are real and widely experienced.

The six objectives defined in Section 6 are each grounded in evidence from at least two of these three sources. This positions THOTH not as a speculative concept, but as a well-justified system with clear technical precedents, demonstrated user demand, and a realistic path to implementation.

9. References

- [1] Velentza, A.-M., Heinke, D., & Wyatt, J. (2020). Museum robot guides or conventional audio guides? An experimental study. *Advanced Robotics*, 34(21–22), 1361–1372.
- [2] Grady, J. et al. (2024). CLIO: A Tour Guide Robot with Co-Speech Actions for Visual Attention Guidance and Enhanced User Engagement. *arXiv preprint arXiv:2512.05389*.